Assignment 2: Recursive Feature Elimination with Linear Regression

github link: https://github.com/andychen910/math-485

Introduction

In this assignment we looked at how Recursive Feature Elimination (RFE) works with linear regression to pick the most useful features. We used the Diabetes dataset from scikit-learn which has 10 features and a target that measures how much diabetes got worse after one year.

Task 1: Dataset Exploration

The dataset has 442 rows and 10 features: age, sex, bmi, bp, s1, s2, s3, s4, s5, and s6. All the values are already normalized so they are all in a similar range. The target goes from 25 to 346 and the average is around 152. I checked for missing values and there were none which made things easier. I split the data 80/20 so 353 samples for training and 89 for testing.

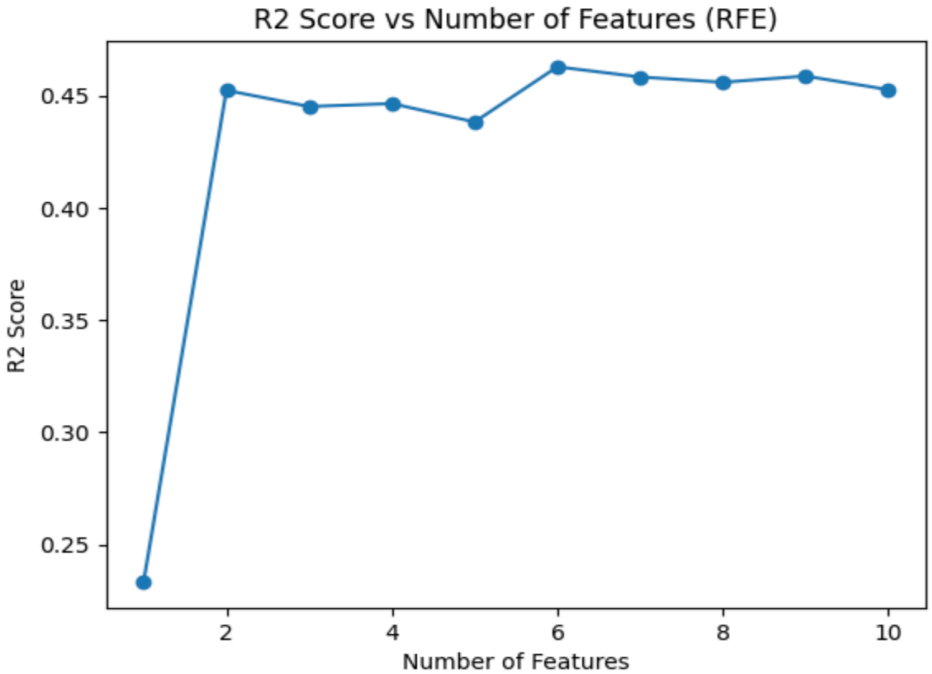
Task 2: Linear Regression Baseline

I trained a basic linear regression model using all 10 features first. The R2 score on the test set came out to 0.4526. This is the starting point I used to compare against the RFE results later.

Task 3: RFE Implementation

I ran RFE by removing one feature at a time starting from 10 features down to 1 and recorded the R2 score each time.

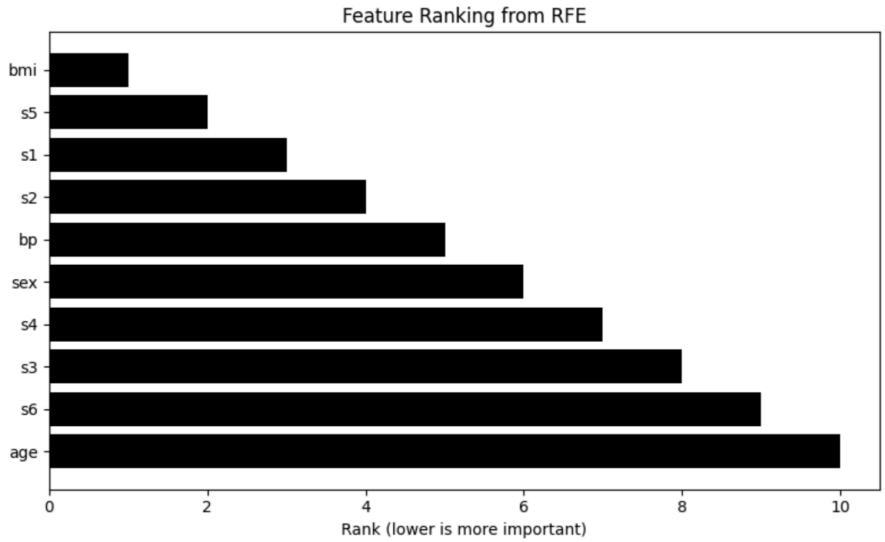
```
R2 score at each step:
  1 feature(s): 0.2334
  2 feature(s): 0.4523
  3 feature(s): 0.4451
  4 feature(s): 0.4464
  5 feature(s): 0.4382
  6 feature(s): 0.4628
  7 feature(s): 0.4583
  8 feature(s): 0.4559
  9 feature(s): 0.4587
  10 feature(s): 0.4526
```



Looking at the plot, 6 features gave the best R2 score of 0.4628 which is actually a bit better than using all 10. So I picked 6 as the optimal number.

Task 4: Feature Importance Analysis

```
Feature ranking (1 = most important, 10 = least important):
Feature  Rank
    bmi     1
     s5     2
     s1     3
     s2     4
     bp     5
    sex     6
     s4     7
     s3     8
     s6     9
    age    10
```



The top 3 features were bmi, s5, and s1. BMI makes sense because its a common thing doctors check for diabetes. s5 is the log of triglycerides which is related to blood sugar. s1 is total cholesterol which also matters for this kind of prediction. The 6 selected features ended up being sex, bmi, bp, s1, s2, and s5. One thing that stood out was that age ranked last which was kind of surprising to me.

Task 5: Reflection

Q1. What did I learn about feature selection using RFE?

I learned that not all features actually help the model predict better. RFE works by removing the least useful feature one at a time, and I could see from the R2 scores that using all 10 features doesnt always give the best result. With just 6 features the model actually did a little better than using all 10. So feature selection is pretty important and can make the model simpler and more accurate at the same time.

Q2. How does RFE compare to LASSO?

LASSO and RFE both help pick useful features but they work differently. LASSO adds a penalty during training so it pushes unimportant coefficients down to zero automatically. RFE is more step by step, it trains the model, removes the weakest feature, then repeats. I think RFE is easier to follow because you can see exactly which features got cut and when. LASSO is faster but its harder to see whats going on inside.

Q3. What insights can you draw about the dataset from the selected features?

The 6 selected features are bmi, bp, s1, s2, s5, and sex. BMI and blood pressure make a lot of sense since doctors always bring those up when talking about diabetes. s5 is related to triglycerides which connects to blood sugar levels. The

biggest surprise was age being ranked last since most people would think age matters a lot for diabetes. Maybe in this dataset the blood test results are just stronger predictors than age alone.